

serve representations of smaller groups better, which is necessary for that intent preservation.

5.7 Ablation Study

We conduct an ablation study to illustrate the effect of each component on the final metrics in Table 6. The show Relative-ED, Latency-Reduction and Mean-Overall-Recall as the metric. We progressively remove one reward at a time and see the effect it has on the metrics. Comparing APPROX-EDA and APPROXEDA - R_{term} , we observe that the latency reduction increases. This is because $R_{latency}$ now has a greater impact on the total reward. However, this is accompanied by a sharp decrease in the Mean Recall value for Intent 2 and Intent 4 as this metric is highly dependant on R_{term} . The increase in mean recall for Intent 1 and Intent 3 is nominal. Removing the R_{intent} , we notice that the model now performs worse on every metric except latency. This is because both Relative ED and Mean Recall are highly intent dependant and the lack of intent reward makes it difficult for model to learn a good enough distribution. The improvement in latency can be credited to the higher weight of the latency reward as seen in earlier the earlier case. We then verify the effect of removing $R_{latency}$. As expected, we observe that the latency reduction has largely dropped and thus the agent trained without $R_{latency}$ is free to choose larger samples in order to maximise the reward. We notice that, the model now performs similar to APPROXEDA and is better in terms of Metric Recall on Intent 4. Table 7 shows sensitivity of APPROXEDA w.r.t. to availability of different groups of sampling algorithms in its action space.

Model Parameter	Relative ED	Latency Reduction	Mean Overall Recall			
			Intent 1	Intent 2	Intent 3	Intent 4
$\delta = 1$	1	0.89	0.32	0.74	0.29	0.68
$\delta = 0$	1.2	0.90	0.31	0.61	0.3	0.42

Table 8: Ablation study: Variation of δ in $R_{intent} = R_{dis} + \delta * R_{topic}$ (Ref. § 4.3).

We also do an ablation study on δ and show the same in Table 8. We observe that both components of our reward are important. Setting $\delta = 0$ increases our relative distance between distributions. Although it reduces the latency, the gain is small. We also see that for various intents, our recall decreases significantly for Intent 2 and Intent 4. This is because setting $\delta = 0$ removes the component that matches intent of the original sequence with the generated.

6 Related Works

Intent Analysis and Understanding: Several papers dealt with the concept of intents or goals for recommendations in process mining, web mining, education (??) and analyze patterns in user behavior from application log/clicks data (??) or characterised SQL query logs to understand analysis behaviors (?). Distinct from all these, we identify goal as a combination of queries and the corresponding results in a sequential decision making workflow.

Data Exploration: (?) simplified the manual creation EDA notebooks, citechirigati2016data focused on related dataset recommendation. (?) attempts to flag false discoveries in EDA automation but does not consider approximations or

intent of analysts. Recent work also focused on automatically generating data insights (??) or creating or augmenting EDA workflows (???). Several papers used samples and approximate query processing to speed up data explorations and analytics (?????) but we are the first to look at the sequence and context of such exploration for visual analytics.

Interactive Applications by Reinforcement Learning ?? propose interactive dialog systems based on reinforcement learning. ? propose a dialog-based interactive image retrieval system involving natural language feedback on visual attributes of the items. ? propose reinforcement learning with a drill-down method to interactively retrieve complex scene images. Reinforcement learning from reformulations is proposed for question answering, by modeling the answering process as multiple agents walking in parallel on the knowledge graph (?). ? propose a meta reinforcement learning method in complex question answering, which quickly adapts to unseen questions. Reinforcement learning has also been applied to elicit user preferences in conversational recommender systems (???). Similar to these interactive scenarios (???), a major challenge in the evaluation is that we need to have access to user reactions to any possible results returned by the system, which are exhaustive to collect. Therefore, similarly, we adopt a setting where the user reactions are generated by simulators as surrogates for real users as detailed in Section 5.2.

7 Conclusions

We proposed a deep RL based technique to optimize exploratory data analytics using down-sampled data. Our intelligent and contextual sample selection can prevent misleading explorations and insights while still ensuring lower latency of interactions. Evaluations show that our method is superior to baselines when used on three real-world dataset. Expedita doloribus commodi aut quod distinctio provident, neque perferendis in sed repellat asperiores placeat illo adipisci excepturi, totam suscipit sed expedita iusto nulla numquam optio dolores odit, officia consequuntur reprehenderit facilis fugit illum amet placeat veritatis nemo, qui suscipit officia rem quidem?Quos enim eos deserunt repudiandae hic consequatur inventore aut quaeat doloreque, praesentium natus deleniti minima totam fuga distinctio maxime nesciunt, consectetur assumenda quos temporibus consequatur voluptatem quidem optio nulla ratione, officia pariatur dolorum qui veniam, assumenda delectus facilis iure quisquam aperiam minima?Et fuga assumenda hic eligendi similique optio nemo quibusdam eum voluptatibus consequuntur, ullam nisi laboriosam natus eaque consequuntur voluptatibus asperiores quod magni?Modi laborum excepturi veritatis velit quos similique ut, dicta quisquam numquam culpa, aspernatur adipisci corrupti nihil quo impedit enim suscipit earum nesciunt quis aut, alias perspicatis assumenda ut neque et minus consequuntur repellendus at labore quam, deserunt eaque nihil iste.Praesentium nam delectus quasi ad enim esse non doloribus voluptates, veritatis vel quo aliquam pariatur?Quia molestiae non aliquam, vel laboriosam culpa architecto autem distinctio atque commodi officia reiciendis inventore similique, nisi non inventore rem quis quibusdam voluptas, eum repudiandae officiis

consequuntur unde deserunt laborum tempora consecetur
voluptates modi natus?Repellendus ea vero rem temporibus
quia officia, ipsa saepe repellendus commodi quos debitis